# Sumedh Kothari

skothari67@gatech.edu | linkedin.com/in/sumedh-kothari | github.com/sumedhk0


## EDUCATION

**Georgia Institute of Technology** **Expected May 2028**
B.S. in Chemical and Biomolecular Engineering
Minor in Computer Science

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout
- **Organizations:** American Institute of Chemical Engineers, Technical Association of the Pulp and Paper Industry

## EXPERIENCE

**Georgia Tech Renewable Bioproducts Institute | Tong Laboratory** Atlanta, GA
*Research Intern* August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, FTIR, and DFT modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

**ChemE Cube | Georgia Tech** Atlanta, GA
*Team Lead - Mechanical* August 2025 - Present

- Led a 5-member mechanical team by overseeing the end-to-end design and CAD of a 1 $ft^3$ DAC system, resulting in a fully integrated prototype from concept through fabrication
- Designed and fabricated DAC hardware using 3D printing and CNC machining to build airflow pathways and zeolite containment resulting in a robust, competition-ready mechanical system
- Engineered the DAC system for a 75% ambient $CO_2$ reduction by optimizing contactor geometry and airflow–sorbent interaction, resulting in a prototype prepared for evaluation at the AIChE national conference (Nov 2026)

**Dressen Laboratory** Los Altos, CA
*Researcher* August 2024 - June 2025

- Designed, tested fluorinated $Mg_2$dobpdc metal organic frameworks for vehicle exhaust $CO_2$ capture, improving capture potential by 20% while improving stability by 115%
- Characterized adsorption efficiency and regeneration performance using TGA, NMR, and cyclic adsorption experiments, with results presented at SYNOPSYS 2025

## PROJECTS

**IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech** Atlanta, GA
*Mechanical/Software Engineer* January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot in SolidWorks by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous navigation and operation with zero human intervention
- Developed and deployed a YOLO-based computer vision model in PyTorch by training and integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identification in landing zones
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles and deploy transponder beacons to certify landing safety

**AiApply - AI Job Applier** Atlanta, GA
*Developer* October 2025

- Developed full-stack AI-powered internship application platform using Python, Selenium WebDriver, HTML, and JavaScript to automate end-to-end job search workflows, including resume optimization, cover letter generation, and programmatic application submission on Handshake
- Implemented Selenium-based automation to extract recruiter profiles from filtered company searches and deliver personalized direct messages at scale, increasing outreach efficiency and response rates
- Engineered web scraping algorithms using Python to identify target companies based on location and industry parameters, then automated tailored email outreach highlighting candidate skillsets parsed from resume data

**GreenGrade - Gemini API Developer Competition** Los Angeles, CA
*Project Lead* July 2024 - August 2025

- Directed 20-person UCLA team to architect and implement a fuzzy logic–driven Sustainability Interval Index calculator with scalable data integration pipelines
- Developed and deployed *GreenGrade* (Flutter/C++), embedding SII computational model and automating sustainability analytics for 175 mining firms through Gemini API

## SKILLS

- **Technical:** Python, Java, Data analysis (Excel, MATLAB), CAD (SolidWorks)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)